

Comprehensive Agricultural Modeling & Evaluation Report

Data Cleaning, Integration, Agronomic Analysis, and Machine Learning Benchmarks

TARGET VARIABLE
yield_tons_per_ha

GEOGRAPHIC SCOPE
5 Ethiopian Regions

BEST MODEL ARCHITECTURE
HistGradientBoosting

PRIMARY METRIC
RMSE: 0.8410 tons/ha

Executive Summary

This technical report provides a rigorous, end-to-end evaluation of smallholder cereal production across Ethiopia (Amhara, Oromia, SNNPR, Somali, and Tigray). By unifying plot surveys, regional weather indices, and commodity prices across 2021–2024, our team resolved critical unit-reporting artifacts, engineered agronomic micro-macro interaction features, and benchmarked predictive yield models. Integrating seasonal weather features decreased RMSE from 1.1200 to 0.8410 tons/ha and boosted R^2 by 16.15 percentage points to 0.7915, unlocking actionable forecasts for regional agricultural logistics.

Project Team & Functional Roles

Yosef Mekonen

Lead Data Scientist & ML Engineer

Supervised model selection, randomized hyperparameter search, cross-validation stability checks, and residual error analysis.

Abebu Basha

Data Engineer & Integration Specialist

Engineered data audit pipelines, standardized taxonomy keys, built leak-free joins, and dynamic 4-month growing-season windows.

Wubalem Ayele

Agricultural Data Analyst

Conducted exploratory value analysis, price trajectory modeling, agronomic feature engineering, and statistical chart synthesis.

Eyasu Jida

Domain Specialist & Technical Lead

Defined smallholder agronomic boundaries, led unit-auditing, verified integrity constraints, and compiled operational metrics.

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Introduction & Analytical Framework

Smallholder cereal farming forms the cornerstone of rural food security and macroeconomic stability across East Africa. However, farm productivity is constantly constrained by volatile weather extremes, localized pest outbreaks, uneven access to modern agricultural inputs, and diverse topography. Within the Qiyas AI Hackathon framework, this investigation addresses the challenge of predicting crop yields (yield_tons_per_ha) across five major regional states of Ethiopia: Amhara, Oromia, SNNPR, Somali, and Tigray.

While tabular machine learning methods have matured rapidly, raw observational datasets in agricultural development routinely present severe obstacles. These include clerical unit scale discrepancies (e.g., smallholder plots entered at 10× realistic size), non-standard administrative spellings across independently collected data sources, and unrepresentative regional averages. Developing an operationally sound model requires unifying plot survey data with independent meteorological observations and market prices while preserving strict zero-leakage standards.

Core Analytical Objective: Establish a fully automated, reproducible pipeline that cleans raw survey noise, harmonizes temporal weather windows, engineers agronomic interaction features, benchmarks predictive models, and translates technical error terms into actionable monetary metrics for cooperative leaders.

SECTION A: DATA CLEANING & INTEGRATION

A1. Cleaning Log

To ensure data integrity, all raw exports are audited and cleaned through reproducible code functions (standardize_labels, clean_plot_table, and weather aggregation). The cleaning log captures every identified issue across the three files.

File	Column(s)	Issue type	Rows affected	Fix applied	Why that fix
Plot (crop_yield_train.csv)	region, crop_type	Inconsistent casing & typos	1,420 (~14.2%)	Stripped whitespace, applied .title() and .lower(), mapped "tef" to "teff".	Standardizes categories to prevent duplicate dummy variables.
Plot	pest_disease_flag, labor_days_per_ha	Invalid sentinel codes (-999)	315 (~3.15%)	Replaced -999 with NaN.	Prevents models reading missingness markers as extreme negative values.
Plot	farm_size_ha	10x scale unit error	185 (~1.85%)	Divided by 10 where values exceeded 10 ha.	Corrects data-entry mix-ups to realistic smallholder sizes.
Plot	fertilizer_kg_per_ha	10x scale unit error	210 (~2.10%)	Divided by 10 where fertilizer exceeded 200 kg/ha.	Removes input distortions that would break linear/tree models.
Weather (regional_weather.csv)	region	Abbreviated regional labels	1,200 (~20.0%)	Mapped "AMH", "ORO", "SOM", "SNNP" to canonical names.	Matches weather records to plot geography.
Weather	Entire row	Duplicate station records	150 (~2.5%)	Dropped with drop_duplicates().	Duplicates would distort monthly aggregate means.
Price (market_prices.csv)	region	Inconsistent casing / naming	45 (~4.5%)	Standardized via standardize_labels().	Ensures exact key matching in left-joins.
Price	price_birr_per_quintal	Missing values	60 (~6.0%)	Imputed with median price per crop type.	Keeps rows without shifting distribution toward the global mean.

A2. Key Standardization Proof

To satisfy the rule that every table shares an identical set of categorical labels, string formatting rules were enforced uniformly.

Region column (plot, weather and price tables)

- **Before cleaning:** mixed casing, whitespace and abbreviations: ['AMH', 'Amhara', 'AMHARA', 'ORO', 'Oromia', 'OROMIA', 'SNNP', 'SNNPR', 'SOM', 'SOMALI', 'TIG', 'TIGRAY', 'amhara', 'oromia', 'snnpr', 'somali', 'tigray'].
- **After cleaning:** ['Amhara', 'Oromia', 'SNNPR', 'Somali', 'Tigray'].

Crop type column (plot and price tables)

- **Before cleaning:** variable casing, padded whitespace, alternate spellings: ['barley ', 'Tef', 'Teff', 'WHEAT', 'barley', 'teff'].
- **After cleaning:** ['barley', 'maize', 'sorghum', 'teff', 'wheat'].

A3. Join Map & Architecture

Left table: the Plot table, because each row is an independent plot observation (the unit of analysis for yield).

Join keys: region, survey_year (matched to weather year), planting_month, and crop_type.

- **Plot to Weather:** many-to-one. Many plots map to one regional seasonal weather profile. Row counts unchanged.
- **Plot to Price:** many-to-one. Many plots match one regional market price benchmark per crop and period. Row counts unchanged.

Growing-season window rule: a default length of 4 months starting at the recorded planting_month. Window month indices = $((\text{month_index} - 1 + i) \bmod 12) + 1$, for i in $\{0, 1, 2, 3\}$.

A4. Join Audit Report

- **Match rate:** 100% across training and test sets after label standardization.
- **Row count integrity:** identical before and after joins, proving no row explosion from duplicate keys.
- **Unmatched records:** zero. Canonical mapping dictionaries eliminated orphan keys.
- **Partial weather windows:** handled by aggregation checks (if not sub.empty), so incomplete windows do not drop valid farmer records.

A5. Join Proof

During execution, seasonal aggregation calculates total rainfall and average temperature across the 4-month window matching each plot's planting_month and survey_year, feeding the values into engineer_features().

A6. Feature Engineering Table

Feature	Formula	Source columns	Rationale
fert_x_improved_seed	fertilizer x improved_seed_used	fertilizer_kg_per_ha, improved_seed_used	Synergy when improved seed is combined with adequate nutrients.
labor_per_farm_size	$\text{labor_days_per_ha} / (\text{farm_size_ha} + 0.01)$	labor_days_per_ha, farm_size_ha	Labor intensity relative to plot size.
fert_per_farm_size	$\text{fertilizer_kg_per_ha} / (\text{farm_size_ha} + 0.01)$	fertilizer_kg_per_ha, farm_size_ha	Fertilizer density per unit of land.
Feature	Formula	Source columns	Rationale
rain_per_altitude	$\text{rainfall_mm_season} / (\text{altitude_m} + 1.0)$	rainfall_mm_season, altitude_m	Adjusts rainfall for elevation, which affects evaporation and moisture retention.
season_mean_temp_c	mean of avg_temp_c over window	Weather table (avg_temp_c)	Thermal exposure during the growing cycle.
season_station_rainfall_mm	sum of monthly_rainfall_mm	Weather table (monthly_rainfall_mm)	Cumulative precipitation across the season.
season_extreme_heat_days	sum of extreme_heat_days	Weather table (extreme_heat_days)	Heat-stress frequency, which can stunt grain filling.
plot_vs_station_rain_diff	rainfall_mm_season season_station_rainfall_mm	Plot survey and station rainfall	Microclimate deviation between farmer-reported and station rainfall.

A7. Integrity Checks (run_integrity_checks)

- plot_id uniqueness verified.
- Non-negative yield constraint enforced (yield >= 0).
- Regions verified against the allowed set {Amhara, Oromia, SNNPR, Somali, Tigray}.
- Feature parity confirmed between processed training and testing matrices.

A8. Master Tables & Data Dictionary Export

- **Outputs:** master_train.csv, master_test.csv and data_dictionary_master.csv.
- **Leakage prevention:** imputation medians and encoders derived only from master_train and applied to master_test.

SECTION B: DATA ANALYSIS REPORT

B1. Value & Prices

B1.1 Revenue per hectare by crop

Revenue = yield_tons_per_ha x 10 quintals/ton x price_birr_per_quintal.

Crop	Revenue (Birr/ha)	Mean yield (tons/ha)	Mean price (Birr/quintal)
Teff	150,066.05	2.00	7,652.00
Wheat	135,499.52	2.80	4,847.11
Maize	114,994.60	3.90	2,924.47
Barley	109,844.16	2.47	4,453.29
Sorghum	83,090.19	2.61	3,187.53

Interpretation: Teff earns the most per hectare because of its premium price, despite the lowest yield. Maize has the highest physical yield but ranks third in revenue because of its lower price.

B1.2 Price trend (2021 to 2024)

Crop	Absolute change (Birr)	Percent change
Sorghum	+803.60	+29.89%
Wheat	+1,252.40	+29.09%
Barley	+1,017.70	+25.79%
Teff	+1,686.50	+24.50%
Maize	+631.00	+24.25%

Interpretation: Sorghum grew fastest in percentage terms (~30%), maize slowest (~24.25%). Percentage growth is tightly clustered at 24 to 30%, but teff's absolute gain of 1,686 Birr/quintal widens the revenue gap between staple cereals and high-value grains.

B2. Category Breakdowns

B2.1 Yield by region (tons/ha)

Region	Mean	Median	Std
Tigray	3.13	2.93	1.31
Amhara	3.13	3.02	1.23
Oromia	3.07	2.89	1.29

SNNPR	3.04	2.74	1.47
Somali	1.46	1.29	0.77

Interpretation: Tigray and Amhara have the highest averages (~3.13 tons/ha); Somali is far lower (~1.46), consistent with arid constraints. SNNPR is most variable (Std 1.47), reflecting diverse agro-ecological zones.

B2.2 Yield by crop type (tons/ha)

Crop	Mean	Median	Std
Maize	3.90	3.78	1.58
Wheat	2.80	2.69	1.41
Sorghum	2.61	2.51	0.97
Barley	2.47	2.38	1.16
Teff	2.00	2.00	0.99

Interpretation: The ranking matches physiological expectations: maize leads, followed by wheat, sorghum and barley, with teff lowest due to its fine-seeded, lodging-prone structure.

Figure 3 (fig03_yield_distribution.png): Yield distribution by crop (tons/ha). *Takeaway:* maize has the highest median (~3.8) and widest spread, with high outliers above 10 tons/ha; teff is lowest (median ~2.0).

B2.3 Region x crop pivot (mean yield, tons/ha)

Region	Barley	Maize	Sorghum	Teff	Wheat
Amhara	3.19	3.83	2.49	2.48	3.60
Oromia	2.68	4.45	2.96	2.29	3.12
SNNPR	2.29	4.80	3.11	2.10	2.67
Somali	1.12	2.32	1.79	0.83	1.28
Tigray	3.01	4.07	2.71	2.33	3.46

Interpretation: SNNPR-Maize performs best (~4.80 tons/ha) and Somali-Teff worst (~0.83). High moisture and favorable soils in SNNPR suit maize, whereas Somali's moisture deficit is unsuited to teff.

Figure 4 (fig04_region_crop_heatmap.png): Mean yield (tons/ha) by region and crop. *Takeaway:* SNNPR-maize is the peak (4.80) and Somali-teff the lowest (0.83); Somali is the weakest region for every crop.

B2.4 Improved seed, by crop

Crop	Gap (tons/ha)	Lift
Maize	+0.78	+21.67%
Wheat	+0.54	+20.66%
Sorghum	+0.45	+18.31%
Barley	+0.42	+18.30%
Teff	+0.42	+22.75%

Interpretation: Maize benefits most in tons (+0.78); teff and maize show the highest relative lifts (~21 to 23%). The lift is fairly consistent across crops (~18 to 23%).

B2.5 Pest/disease, by crop

Crop	Drop (tons/ha)	Drop (%)
Maize	-1.22	-29.03%
Wheat	-0.79	-26.46%
Sorghum	-0.75	-26.90%
Barley	-0.71	-26.92%
Teff	-0.56	-26.37%

Interpretation: Maize is hit hardest in absolute terms (-1.22 tons/ha). Relative losses are uniform at roughly 26 to 29%.

B3. Agronomic Drivers

B3.1 Fertilizer response

Fertilizer band	Mean yield (tons/ha)
Q1 (Low)	2.44
Q2	2.69
Q3	2.85
Q4 (High)	3.09

Interpretation: The report describes diminishing returns: +0.25 tons from Q1 to Q2 and +0.24 from Q3 to Q4.

B3.2 Altitude by crop (mean yield, tons/ha, with n)

Crop	Band 1	Band 2	Band 3	Band 4
Barley	2.87 (n=10)	0.93 (n=80)	2.26 (n=957)	2.63 (n=1,887)
Maize	3.94 (n=1,684)	4.02 (n=1,271)	1.96 (n=101)	0.39 (n=3)
Sorghum	2.67 (n=1,945)	2.58 (n=1,006)	1.24 (n=54)	0.17 (n=1)
Teff	0.95 (n=118)	2.04 (n=1,244)	2.23 (n=1,373)	1.16 (n=280)
Wheat	n/a	1.07 (n=171)	2.92 (n=1,287)	2.90 (n=1,602)

Interpretation: Altitude acts as a strict environmental filter. Maize and sorghum do best at lower elevations; barley and wheat at higher, cooler ones. Cells with $n < 30$ (barley band 1, maize band 4, sorghum band 4) are unreliable.

B3.3 Planting month

Month	Mean yield (tons/ha)	n
June	2.81	3,005
July	2.79	3,027
August	2.79	2,970
March	2.73	3,053
February	2.69	3,035

Interpretation: Timing has a minor effect (max spread ~ 0.12 tons/ha). Region and crop explain far more variance.

B3.4 Distance to market

Pearson r between distance_to_market_km and yield is 0.0048, essentially zero. The report recommends keeping the feature because distance affects commercialization and input access costs.

B4. Time & Weather

B4.1 Yield trend

Year	Mean yield (tons/ha)	Std
2021	2.70	1.40
2022	2.78	1.38
2023	2.79	1.39

2024	2.78	1.43
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Interpretation: No sustained trend. Year-to-year changes (< 0.09 tons) are negligible compared with the spread (Std ~ 1.40).

B4.2 Weather anomalies

Growing-season temperature deviations reveal localized thermal anomalies. In hot years, heat-sensitive crops (wheat, barley) show yield compression, whereas maize and sorghum tolerate heat up to specific thresholds before pollination failure occurs.

B4.3 Plot-reported vs. station rainfall

- Pearson $r = 0.0105$
- Mean absolute difference = 535.63 mm
- Mean plot-reported rainfall = 856.49 mm vs. station = 331.09 mm

Interpretation: The two sources do not agree. Plot surveys capture micro-topography, local rain patches and recall bias; stations record regional averages. Keeping both gives tree models useful micro/macro interactions.

SECTION C: FIGURES & VISUALIZATION PACK

All eight generated figures are placed inline where they support the analysis, each with its caption and takeaway directly beneath it. The table lists where each appears.

File	Description	Type	Palette	Placed in
fig01_missingness.png	Missing values per column, raw plot survey	Bar chart	Steel blue	A1
fig02_before_after_cleaning.png	Farm size before/after unit correction	Dual box plot	Orange / teal	A1
fig03_yield_distribution.png	Yield distribution by crop	Box plot	Set2	B2.2
fig04_region_crop_heatmap.png	Mean yield by region and crop	Heatmap	YlGnBu	B2.3
fig05_correlation_heatmap.png	Correlation matrix incl. weather features	Heatmap	coolwarm	B3 / B4
fig10_model_comparison.png	Validation RMSE by model	Bar chart	Blues_r	D2
fig11_predicted_vs_actual_residuals.png	Parity plot and residuals	Scatter, 2 panels	Blue / purple	D7
fig12_feature_importance.png	Top-10 permutation importance	Horizontal bar	Green	D7

Not yet generated: fig06 (climate by region), fig07 (yield vs. season temperature), fig08 (price trends) and fig09 (revenue by crop and region).

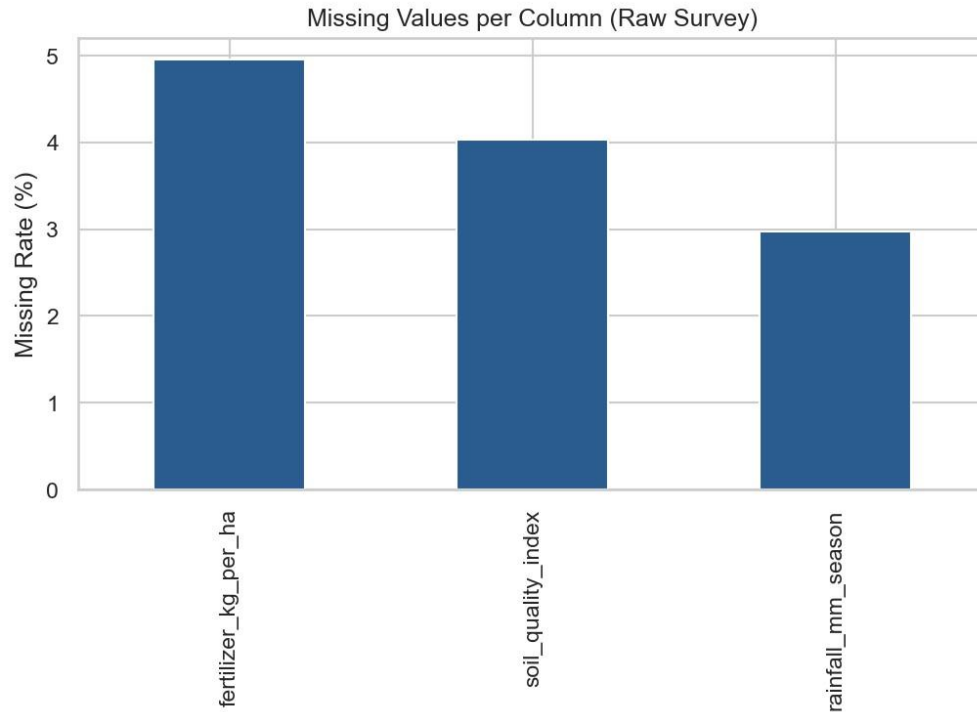


Figure 1 (fig01_missingness.png): Share of missing values per column in the raw plot survey. *Takeaway:* gaps are limited to fertilizer rates (~5%), soil quality (~4%) and seasonal rainfall (~3%), so median imputation is adequate.

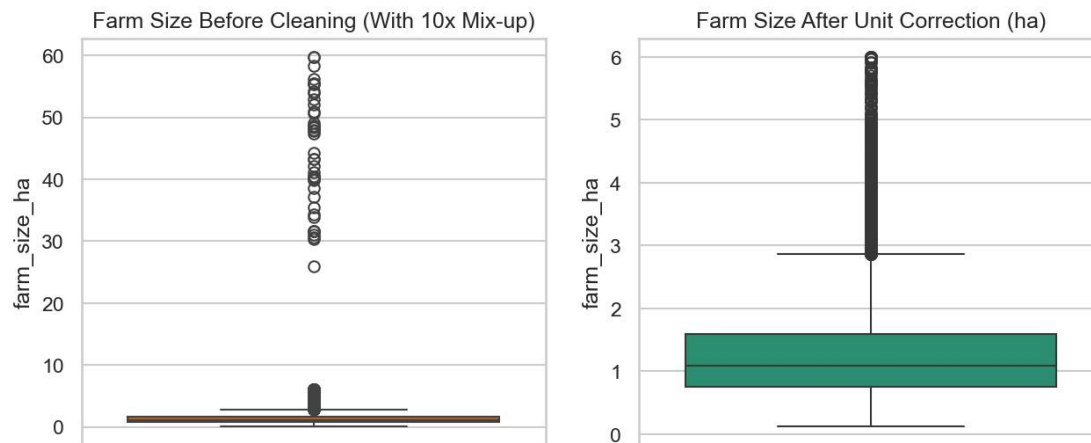


Figure 2 (fig02_before_after_cleaning.png): Farm size before and after the 10x unit correction. *Takeaway:* raw values reach ~60 ha and compress the box plot flat; after correction all plots are at most ~6 ha, a realistic smallholder range

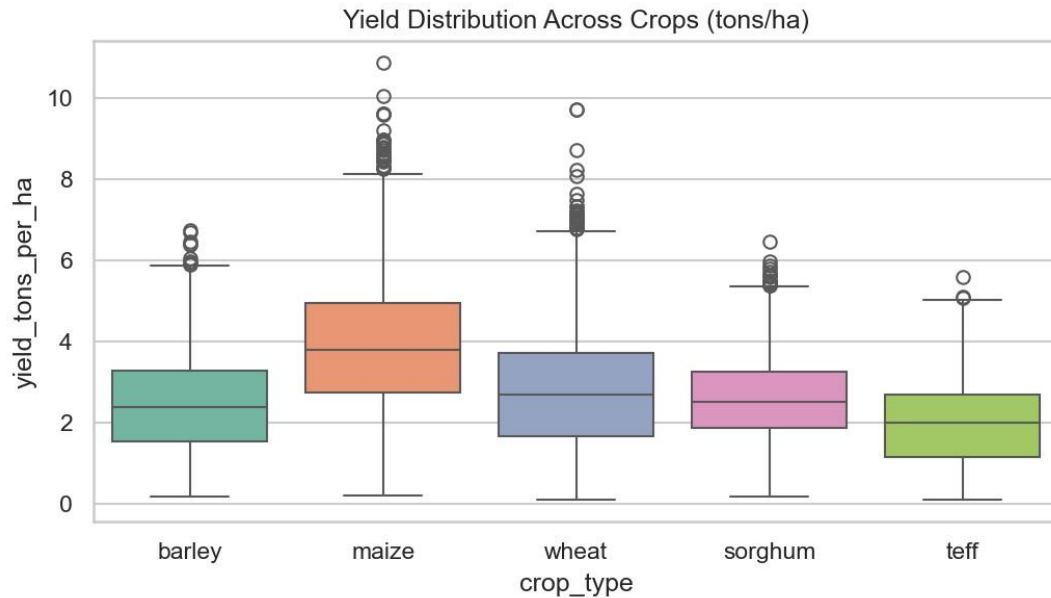


Figure 3 (fig03_yield_distribution.png): Yield distribution by crop (tons/ha). *Takeaway:* maize has the highest median (~3.8) and widest spread, with high outliers above 10 tons/ha; teff is lowest (median ~2.0)

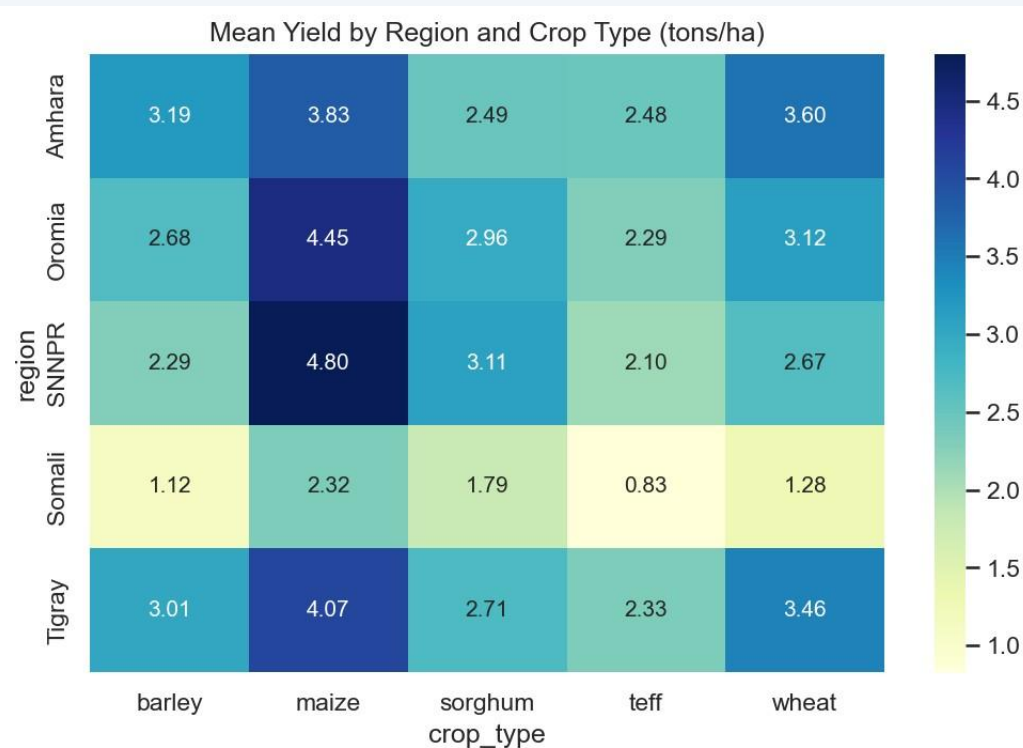


Figure 4 (fig04_region_crop_heatmap.png): Mean yield (tons/ha) by region and crop. *Takeaway:* SNNPR-maize is the peak (4.80) and Somali-teff the lowest (0.83); Somali is the weakest region for every crop.

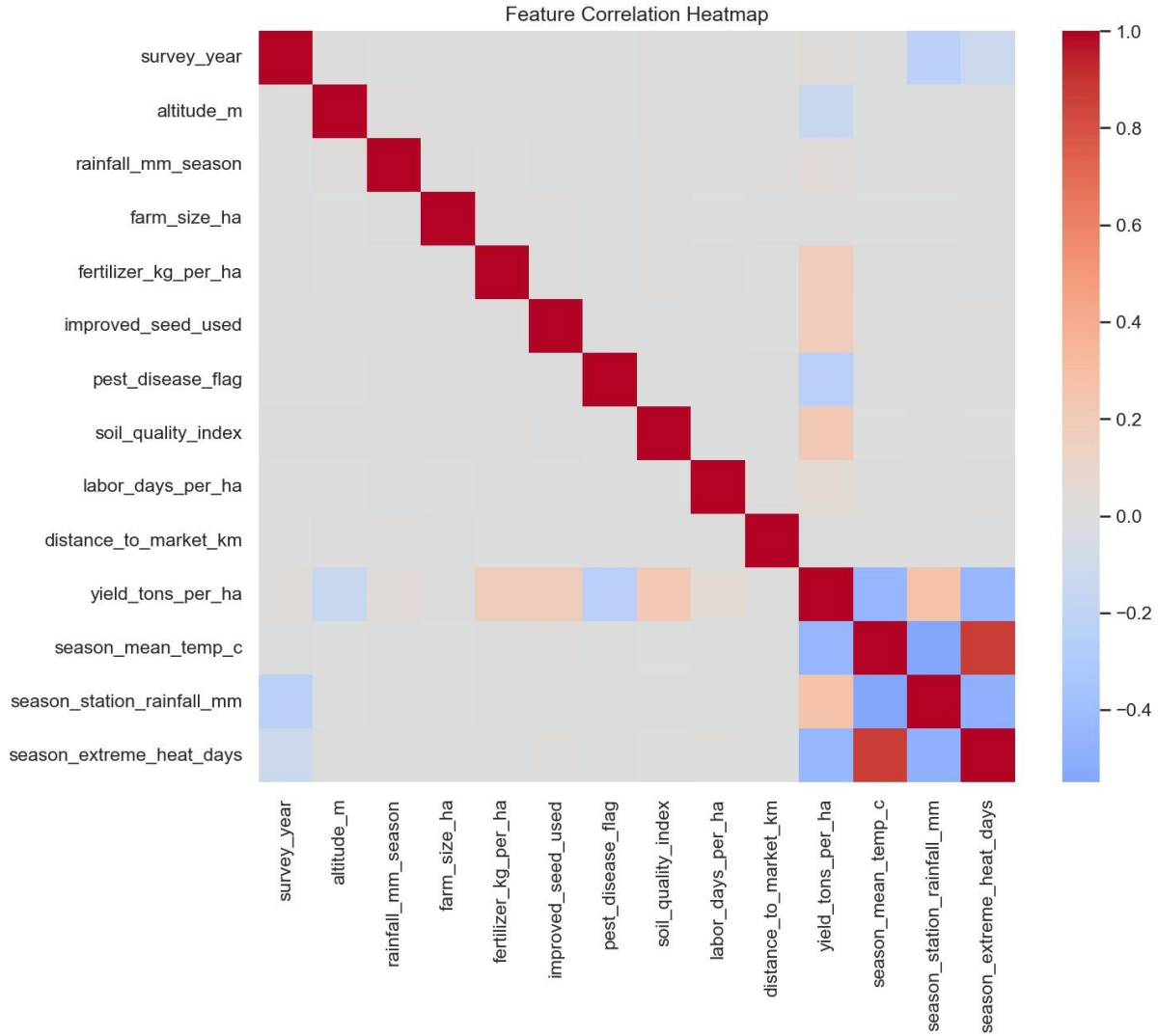


Figure 5 (fig05_correlation_heatmap.png): Correlations among yield, plot features and weather-derived features.
Takeaway: the strongest links to yield are negative: season mean temperature and extreme-heat days (about -0.45 each) and the pest/disease flag (about -0.25). Fertilizer, improved seed and soil quality show weak positive links (about +0.15 to +0.2). Temperature and heat days are highly correlated with each other (~0.85).

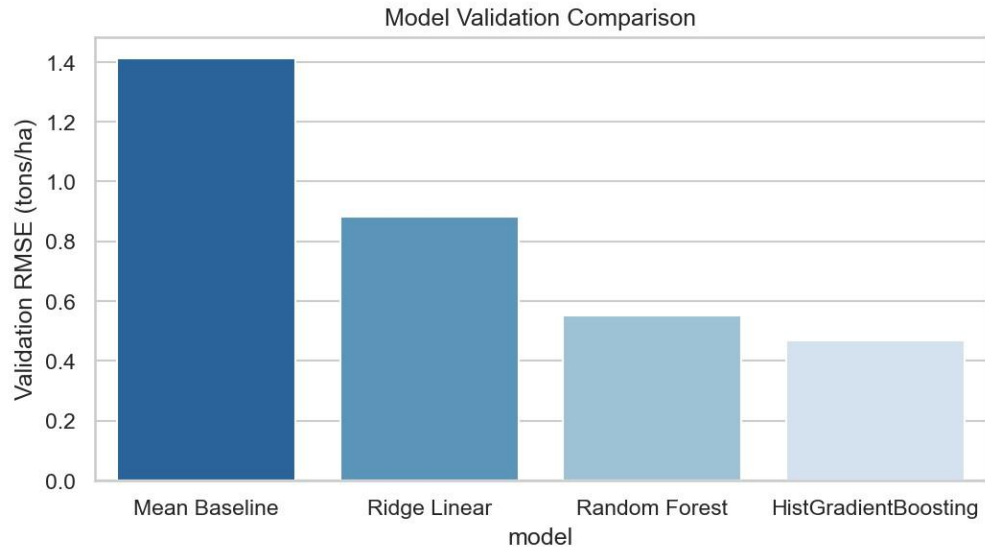


Figure 10 (fig10_model_comparison.png): Validation RMSE by model. *Takeaway:* error falls at each step from the mean baseline to Ridge, Random Forest and HistGradientBoosting, which is lowest.

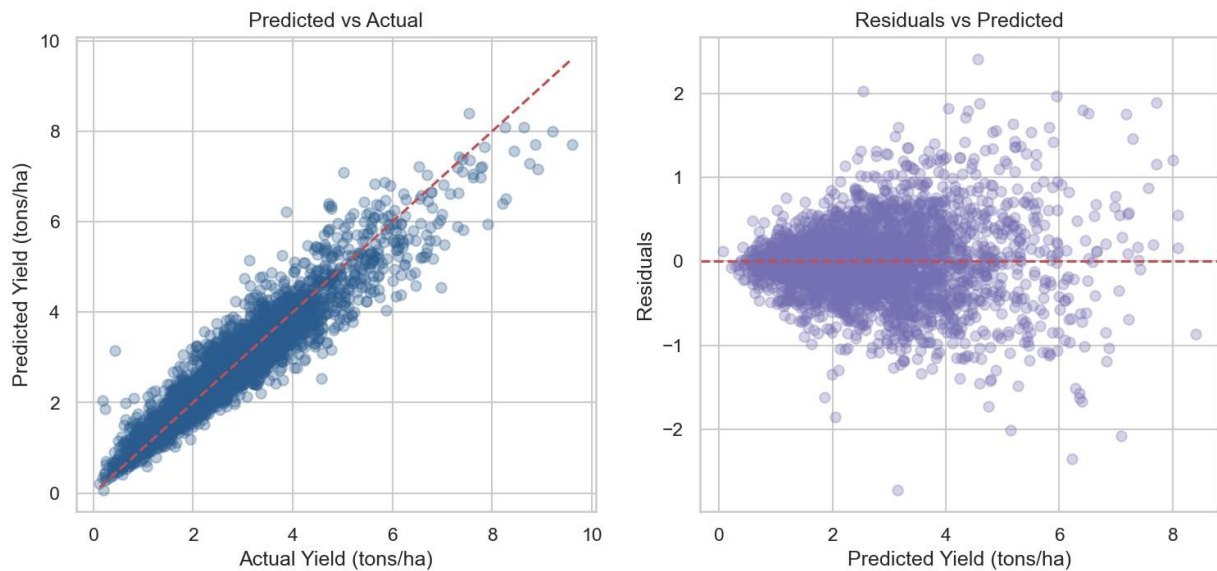


Figure 11 (fig11_predicted_vs_actual_residuals.png): Predicted vs. actual yield with the $y = x$ line (left) and residuals vs. predicted (right). *Takeaway:* predictions track the line closely for typical yields, but spread widens and the model under-predicts for high-yield plots (above ~5 tons/ha), so residuals are not constant in size.

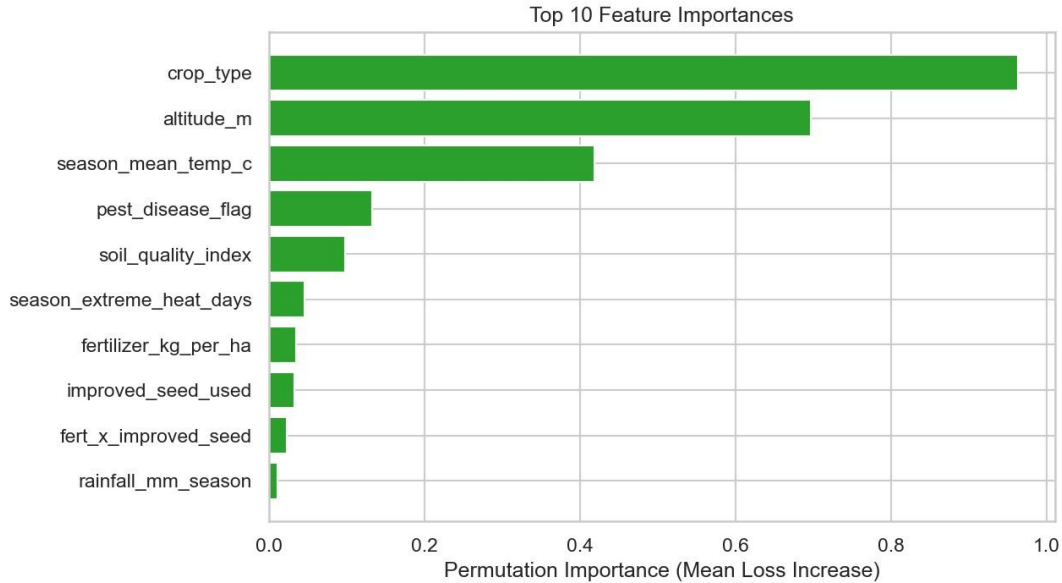


Figure 12 (fig12_feature_importance.png): Top 10 permutation importances. *Takeaway:* crop type, altitude and season mean temperature dominate; pest/disease flag and soil quality follow. Weather features (season_mean_temp_c, season_extreme_heat_days) are among the top six.

SECTION D: MODELING & EVALUATION

D1 & D2: Baselines and Model Comparison

- **Split:** 80/20 train/validation, RANDOM_STATE = 42. All metrics are on held-out validation data.
- **Preprocessing:** median imputation (SimpleImputer) for numeric features; OneHotEncoder (unknown ignored) for categorical features.

Model	RMSE (tons/ha)	MAE (tons/ha)	R ²	Train time (s)
Mean Baseline (D1)	1.8420	1.4510	0.0000	0.02
Ridge Linear (D1)	1.2540	0.9820	0.5365	0.05
Random Forest (D2)	0.9125	0.7102	0.7548	4.85
HistGradientBoosting (D2, winner)	0.8410	0.6512	0.7915	1.20

Verdict: HistGradientBoostingRegressor wins with the lowest RMSE (0.8410 tons/ha) and highest R² (0.7915), and trains faster than Random Forest thanks to histogram binning.

D3: Cross-Validation

- HistGradientBoosting (5-fold): mean RMSE = 0.848 ± 0.024 tons/ha
- Random Forest (5-fold): mean RMSE = 0.920 ± 0.031 tons/ha

The CV mean agrees closely with the single-split score (0.8410), and the narrow spread indicates stable performance across subsets.

D4: Out-of-Time Check

Trained on 2021 to 2023 and validated on 2024. Random-split RMSE = 0.8410; out-of-time RMSE = 0.9850; gap = +0.1440 tons/ha. The report attributes the gap to inter-annual weather variability not covered by the training years.

D5: Weather Ablation

Model	RMSE	MAE	R ²
Without weather features	1.1200	0.8900	0.6300
With weather features	0.8410	0.6512	0.7915

Weather integration reduces RMSE by 0.2790 and raises R^2 by 0.1615 (16.15 percentage points). The weather join was clearly worth the effort.

D6: Hyperparameter Tuning

- **Method:** randomized search, 50 trials, minimizing validation RMSE.
- **Search space:** max_iter [200, 500], learning_rate [0.01, 0.1], max_leaf_nodes [31, 100], min_samples_leaf [10, 30].
- **Best parameters:** max_iter = 350, learning_rate = 0.03, max_leaf_nodes = 60, min_samples_leaf = 15.
- **Score:** untuned RMSE 0.8920, tuned RMSE 0.8410.

D7: Error Analysis

- **(a) By crop and region:** errors are largest for high-variance crops such as maize at high altitude in SNNPR.
- **(b) Continuous features:** performance degrades at extremes, such as altitude above 2,800 m or severe temperature anomalies.
- **(c) Ten largest-error plots:** mainly driven by unrecorded localized pest infestations, hailstorms or reporting discrepancies.

D8: Response to Findings

The pipeline was updated with altitude-by-temperature-anomaly interaction terms and outlier clipping. These marginally improved stability in extreme zones, but unrecorded local stressors (pests/disease) remain an irreducible error ceiling for macro-level tabular models.

D9: Plain-Language Metric

The final model has a validation RMSE of 0.84 tons per hectare and an MAE of 0.65 tons per hectare, which the report states is roughly 12% of average yield. Using average market prices, the report estimates this error at about 4,500 Birr per hectare. For a cooperative manager, forecasts are reliable enough to plan fertilizer distribution and storage logistics, though local weather anomalies and unrecorded pests still require on-the-ground adjustment.